

**Question 1**

The beam AB shown in Figure Q1 is simply supported by a pin at A and a roller at B. The beam has a span of 4.0 m and carries a single vertical point load of 20 kN located 1.5 m from support A. The self-weight of the beam may be ignored.

- (a) Determine the vertical support reactions at A and B.
- (b) Draw the shear force diagram, showing all significant values.
- (c) Draw the bending moment diagram, stating the maximum bending moment and its position.
- (d) Sketch the deflected shape of the beam.

*Figure Q1*